



Beer distribution isn't trivial

Damm operates **~200 trucks per day** delivering to thousands of bars, restaurants, and supermarkets across Spain, and each route a complex puzzle of stops, mixed cargo, and tight capacity constraints.

Scale

Hundreds of stops daily, spanning every region of Spain

Complexity

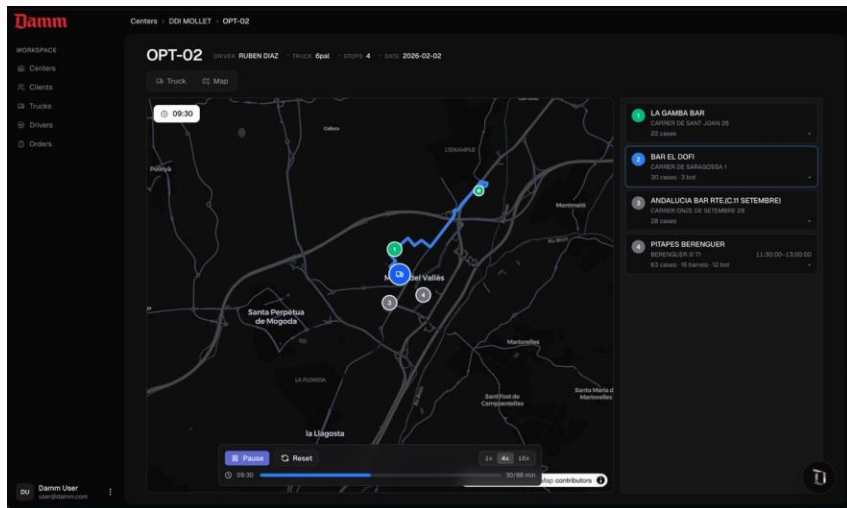
Mixed loads — bottle cases, cans, kegs — all in one truck

The Gap

Load planning is still largely **manual**: slow, error-prone, and costly

Our Solution

One screen to see and validate the entire route



A **Next.js + React** web application that unifies every planning decision into a single, coherent interface — before the truck ever leaves the depot.

Interactive Route Map

All stops plotted and ordered at a glance

Ordered Customer List

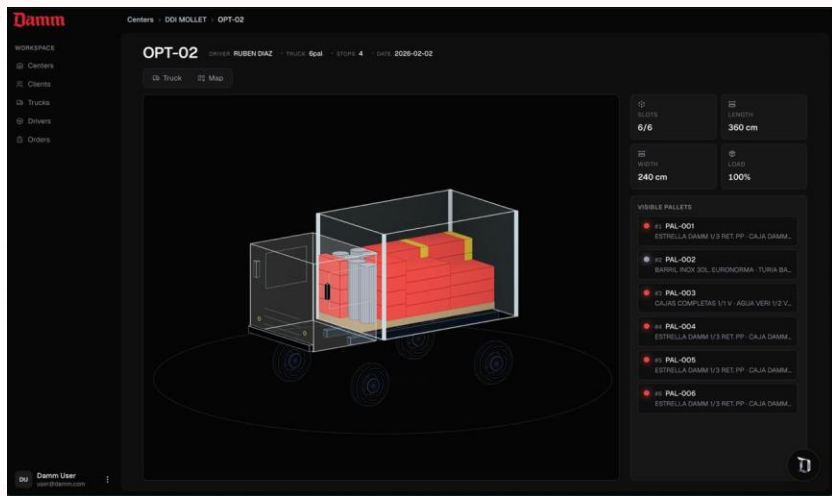
Sequence-aware stop management in one panel

3D Truck Visualizer

Inspect any pallet with a single click

Hero Feature

The truck, pallet by pallet



An iso-view **3D wireframe** of the real Damm truck, 6-pallet, 8-pallet, or van: rendered with **Three.js / React Three Fiber**.

i Hover any pallet → sidebar highlights it. Click → inline panel shows customer, SKU, weight & volume. Fully keyboard accessible.

■ Bottle Cases

Red: standard rectangular pallets

■ Can Cases

Blue: lighter, distinct profile

■ Kegs

Grey: 2×2 cylinder footprint

Optimization + contract, ready to scale

How It Works Today

Python backend with an optimization engine handling multi-customer pallet bin-packing and multi CVRPTW problem solvers using Google OR-Tools through constraint optimization modelling.

A **typed API contract**: decouples the optimization engine from the render layer entirely. Evolve one without touching the other. Having a clean and scalable architecture.

Next Steps

01

Live Integration

Wire Damm's real load-plan endpoint directly into the app

02

Enhanced returnable predictions

Handle returnable packaging in the load plan using predictive models

03

Fleet Analytics

Fleet-wide utilization metrics and efficiency reporting